

Streamlining Epilepsy Surgery Planning Rounds with Virtual Reality

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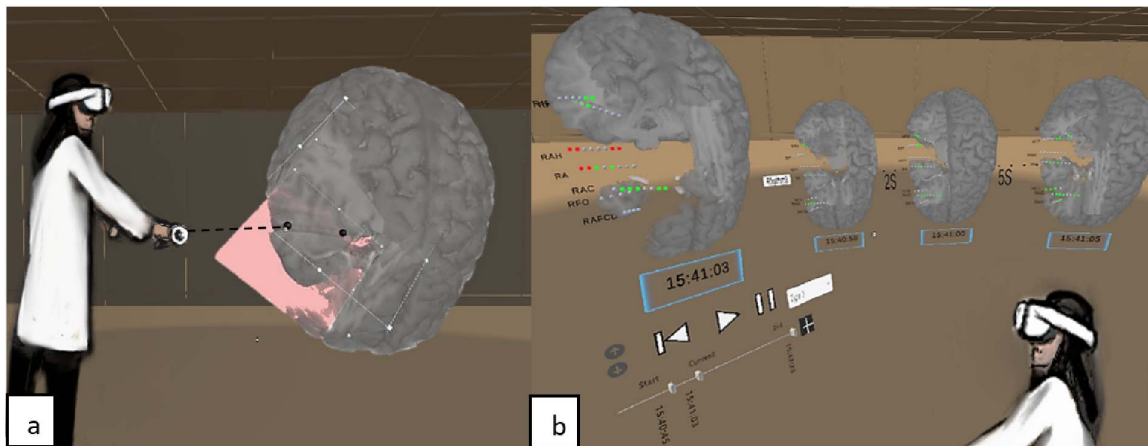


Figure 1: Using our virtual reality (VR) interface for data preparation and presentation in an epilepsy surgical planning round. a) In the preparation step, a neurologist can mark electrodes' endpoints in a 3D brain using VR interactions. b) The system visualizes seizure propagation with different types in time intervals for presentation.

ABSTRACT

We introduce RealityFlow, a virtual reality (VR) system designed to assist neurologists in a clinical workflow of planning epilepsy surgery. RealityFlow offers 3D direct VR manipulation to prepare data, integrates spatial and temporal information pieces for demonstration in a 3D space, and introduces a novel visualization of seizure propagation to help neurologists analyze and present a seizure spread and its types.

Index Terms: Human-centered computing—Virtual reality; Human-centered computing—Scientific visualization; Human-centered computing—Visualization—Visualization techniques;

1 INTRODUCTION

In this paper, we present an approach of using virtual reality (VR) to support neurology teams in planning brain surgery for the treatment of epilepsy. VR can potentially improve data exploration and demonstration of planning and performing neurosurgery [6]. VR is an immersive technology that can enhance the understanding of

brain anatomy and provide natural interactions with a 3D brain [2]. A user can place neuroscience information layouts in a VR environment, reducing the required physical space for setting up multiple physical monitors in working areas [11]. Head-mounted VR devices (HMD) provide a fast change of view in a VR space. Our HMD VR tool aims to aid neurologists in preparing their analyses and presenting the results of their investigation in a medical meeting for epilepsy surgical decision-making (Fig. 1).

At two primary hospitals in our province, a small group of neurologists focuses on diagnosing and treating epilepsy. About one-third of epileptic patients require surgery, of which about 60% are seizure-free after surgery [10]. Accurate surgical diagnosis is crucial for positive surgical outcomes. Improvements in diagnosis can possibly increase the chances of post-surgery seizure freedom.

Seizure spread characterizes when and where a seizure starts, propagates, and terminates inside a brain and is crucial information for surgical planning. Seizures may last minutes and recur several times when a patient is observed in a hospital seizure monitoring unit. Seizures may have different rhythmic types, with unique seizure spread patterns in the recording period. To determine which parts of a brain should be resected to reduce seizure occurrence, neurologists need to understand where seizures start in a brain and how they propagate through it. Also, they identify and compare different seizure rhythmic types in the brain. To do this, they implant electrodes into the brain that measure electrical activity over time at specific locations.

Neurologists use magnetic resonance imaging (MRI) scans to locate these electrodes and their contact points in a brain (spatial information), and they use time series data of Stereoelectroencephalog-

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raphy EEG electrodes (sEEG) to understand electrical activity over time (temporal data) at specific brain locations. Neurologists need to mentally integrate these two data sources of a patient's clinical data to understand how seizures spread. RealityFlow generates visualizations that help with the mental integration of the collected spatio-temporal data. The surgical planning workflow includes two steps.

The first step is the analysis of seizure spread. In current practice, neurologists localize implanted electrodes by studying 2D planes of an MRI brain scan on a 2D screen, commonly using the Horos software tool¹. Then, they analyze sEEG seizure activity, usually in a software tool called Natus². They mentally map the locations of implanted electrodes and seizure recordings and identify seizure onset (starting zone). Then, they prepare their analysis results for a presentation at the Comprehensive Epilepsy Program (CEP) meeting. Neurologists rely on personal approaches to prepare their investigation results for CEP presentations [3]. They search through 2D plane images of MRI to find suitable views of target electrodes to present and write down spread information for demonstration. They might take and prepare several screenshots from MRI 2D images to show a proper view of a single electrode. Others might present their markings (annotations) on seizures using Natus. The procedure is called the EEG reading phase.

The second step is presenting their analysis and eliciting other specialists' feedback. In surgical planning rounds, physicians evaluate the chances of seizure freedom when performing surgery and a potential lesion (damaged tissue) area to be resected from a brain. Meeting audiences give feedback on possible further operations to be performed on patients. A physician will present each patient case in a half-hour session. The audience imagines 3D brain anatomy and 4D seizure spread for decision-making. A neurologist presenter usually switches back and forth between different software tools to show patient data. Several iterations of EEG reading phases and presentation periods might be required to reach clinical consensus before suggesting surgery.

The pre-surgical planning workflow is intensive, time-consuming, and imposes a high mental load on the small community of neurologists with crucial roles in surgical decision-making as it requires understanding a 4D seizure spread. Deploying new diagnostic tools and effective communications are among the suggested interventional strategies for improving epilepsy surgical workflow [16]. Initial positive feedback suggests VR can be beneficial for visualizing seizure propagation [2]. RealityFlow, proposing a novel visualization technique, aims to help neurologists in the clinical workflow (Fig 1).

Animation and small multiples are the two most popular techniques for spatiotemporal data visualization [1]. In animation, time is mapped to time; in small multiples, time is mapped to space. Several studies have shown that each of the two techniques is suitable for performing specific sets of tasks [1, 15]. Animation can show trends and subtle changes through time, although tracking changes in the animation visualization might be hard for users and can cause cognitive loads for comparing several time points. Commonly, the animation is user-preferred, and using animation can be suitable for cases when a presenter explains data and guides attention through the presentation [15]. Using small multiples is suggested to be used for comparison instead of animation [15]. Small multiples are suitable for comparison, but enough screen is needed to arrange different multiples in a display space [1]. It is hard to perform a detailed inspection and compare time windows over a long-time duration using traditional animation or small multiples [4, 12]. Users need to remember different time points using the animation technique for long intervals, or they would require interacting with small multiples back and forth to find suitable intervals [12].

¹<https://horosproject.org/>

²<https://natus.com/>

Gleicher [7] suggests combining different comparative visualization techniques and interactions to prepare a complete concept with scalability. In a large empty physical room where a user can freely move, the VR provides space for assembling and combining the visualization components. In the RealityFlow visualization, we combine positive characteristics of small multiples and animation to create views that support the comparisons of seizure rhythms in different time intervals and highlight sudden seizure changes. Our technique is scalable to visualize a week-long of recording in a hospital.

Our contributions include the following:

- A new visualization for seizure spread that combines small multiples and animation to support visual comparison of long-term seizure monitoring data
- A new prototype tool, RealityFlow, that uses these visualizations to support data analysis, preparation, and presentation of seizure spread data in epilepsy presurgical evaluation rounds

2 RELATED WORK

We discuss related work for the visualization of spatiotemporal visualization and studies focusing on epilepsy presurgical evaluation.

2.1 Animation and Small Multiples

Reducing the visualization space for arranging time-varying layouts is an approach for efficient sequential time scanning and comparison. Inspired by small multiples, time intervals can be summarized into vignettes (multiples) [12] or spatially organized and grouped into piles [4]. Lee et al. [12] proposed an interactive visualization technique known as time slicing, which accelerates a comparison of network changes by selecting different periods. Each vignette [12] shows a network summary in a time interval. The time-slicing method does not show the dynamics of changes using animation. Bach et al. [4] focused on aggregating similar objects into piles decreasing the required 2D screen for comparing objects by summarization. A user should interact with each data display one by one to preview the changes in each pile, which could be intensive.

Virtual reality can provide enough space for the visualization of animation and small multiples or their integration in a large physical space [5, 13]. VR provided enough space for FiberClay [8] to position small multiples on a floor and use animation to show transitions between multiples of neural activity instances and to demonstrate a dynamic flow of air traffic. BentoBox [9] juxtaposes time-varying instances of scientific data and shows organs with different characteristics using small multiples and animation. One of the features of BentoBox is to position snapshots of the body organ visualization and enable a user to interact with animation to improve comparison. Users can scrub through an animated visualization and add additional layouts to compare static time points side by side. However, comparing time points and adding layouts for several intervals to choose each time point one by one might be effortful.

2.2 Epilepsy Pre-surgical Evaluation

EpiNav™ [18] is the most similar software tool to our VR system, helping with sEEG signal visualization and resection planning. In a desktop-based mode, EpiNav requires three different images of pre-operation MRI, post-operation computed tomography scan, and parcellation images (the segmentation of brain anatomy) to automatically localize electrodes based on thresholds on image pixel values and the position of bolts (end of electrodes). A user can investigate a brain by rotating or changing a brain's surface (outer region) using a 3D view of the brain. To browse the inner region of a brain, a user must interact with 2D planes in a desktop-based application. This interaction technique can have the same problem as Horos for interaction with a brain, which can be laborious. A user can view seizure propagation with animation, which can be

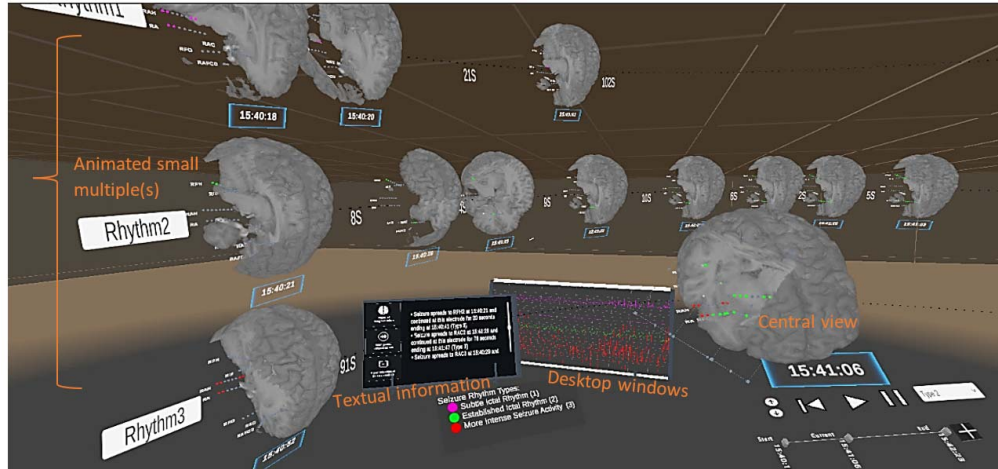


Figure 2: Users can analyze and demonstrate seizure propagation in RealityFlow. The tool includes different information layouts.

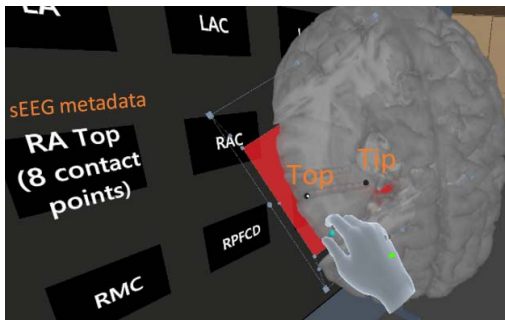


Figure 3: Marking the tip and top (endpoints) of an electrode (RA) in a brain using a cutting cube.

intensive to explore, especially for longer recording periods. Also, A user indirectly interacts with animation by moving the EEG slider. This software does not visualize different seizure types. EpiNav also shows only one view of a brain at a time in animation.

Immersive technologies provide natural and direct interactions to analyze spatial data visualization, including volumetric data. VR-RRRoom [17] is a virtual reality system that enables radiologists to interact directly with volumetric data using a multi-touch surface. For epilepsy presurgical evaluation [2], Aminolroaya et al. [2] focused on the visualization of a seizure spread inside a brain. However, different seizure types are not explicitly arranged for comparison. Users should use an animation temporal slider or play the animation video back and forth to compare different intervals of seizure types. This can be effortful for a long interval investigation.

3 REALITYFLOW

In this section, we describe our prototype, RealityFlow (Fig. 2).

3.1 Overall Rationale (Design Choices)

RealityFlow aims to help neurologists in the epilepsy pre-surgical evaluation procedure. Neurologists can mark electrodes that record seizures inside a brain using RealityFlow. Instead of desktop-based exploration of 2D MRI images, users can use direct interactions with VR controllers to manipulate a 3D brain. RealityFlow receives seizures' temporal information, such as seizure start, end, and types, by processing output texts (annotations) from Natus.

Then, RealityFlow integrates spatial and temporal information pieces of the seizure's spread. Rather than neurologists imagining seizure propagation in a brain and writing down the spread information, RealityFlow visualizes seizure spread in a virtual brain and also provides the spread information in a text format (Fig. 2). Neurologists also can explore and compare seizure types in different time intervals. Instead of back-and-forth investigations of long recording time windows for comparing seizure propagation types, neurologists can select and analyze the visualization of the different types in several layouts of the VR space. The domain experts can summarize their analysis for presentation in the CEP meetings. As opposed to taking several 2D pictures of MRI data, they can prepare and present different views of an electrode on the fly. Neurologists can also move the main camera to view positioned layouts without any back-and-forth switching between software tools. Finally, the medical experts can share and update their analyses based on discussions in the CEP cycles. The following parts discuss how RealityFlow receives spatial and temporal input as well as how it visualizes seizure spread's spatio-temporal information.

3.2 Spatial data

RealityFlow enables neurologists to analyze and manipulate a 3D representation of a brain MRI scan to mark electrodes inside the brain. First, RealityFlow generates a visualization of a real patient's 3D brain tissue. We use the BrainSuite³ service for removing the brain scalp from our MRI images. To help neurologists to get information about each electrode before marking it, our VR tool extracts the number of contact points in each electrode from sEEG file metadata and shows users this information (Fig. 3). Each electrode has several small contact points.

Medical experts can manipulate a cutting cube and the brain to analyze brain tissue and mark electrodes inside a 3D brain in VR (Fig. 3). The tool allows neurologists to rotate, move, and scale the brain and the cutting cube. Users select two end points of an electrode. Then, our tool will generate the contact points of the electrode based on sEEG metadata and marked endpoints.

3.3 Temporal data

The second step is integrating temporal information about seizure activities based on sEEG data. Our system receives a seizure's temporal information prepared by neurologists using the EEG analysis

³<http://brainsuite.org/>

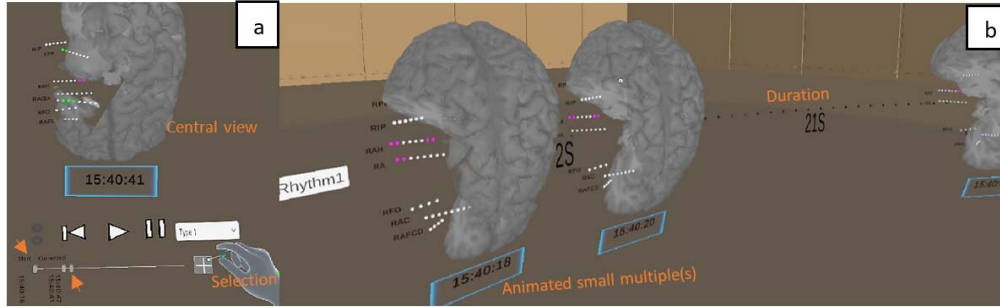


Figure 4: a) The user can select an interval of a seizure rhythm (type 1) in the central view to be shown by the animated small multiples. b) The animated small multiples layout shows the user-selected interval of the seizure type in a row.

software, Natus. Currently, users annotate brain activities by entering information about seizures and identifying the duration of seizures in Natus. RealityFlow receives these annotations from Natus in a text file. The annotated data includes information about seizure onset, seizure propagation, termination, and types. The system processes the text to extract important information using a rule-based natural language processing (NLP) approach [14]. First, the system preprocesses data, such as lower casing words and removing extra spaces in each line. Then, the algorithm looks for specific keywords (like “onset” and “fast”) to identify the start and end of seizures, durations, and seizure types.

Based on spatial and temporal information pieces, RealityFlow visualizes seizure spread. RealityFlow also mirrors desktop windows so that users can also view Natus or any other medical software tools from within the VR system (Fig. 2). Our system summarizes neurologist analyses in graphical and textual formats (Fig. 2). Neurologists can also save and retrieve their analyses for presentation.

3.4 Seizure propagation visualization

RealityFlow also visualizes seizure propagation and its types. Our visualization includes two visual components. We call RealityFlow’s first visual component a *central view*. The central view is a visualization of a 3D brain. Users can analyze the dynamics of sudden changes in the seizure spread and its types using an interactive animation in the central view. Neurologists can interact with the central view using a timeline slider and a media controller (Fig. 4(a)). They can change start and end times to focus on a specific time interval, showing changes over a user-selected time interval. For example, a neurologist can select and investigate an animated visualization of a seizure propagation in a thirty-second time interval instead of a week of recording.

Neurologists can select and organize time intervals from a central view to create the animated small multiples layout (Fig. 4(b)). Inspired by small multiples and animation, we provided animated small multiples to accelerate the comparison task of seizure types by arranging layouts and to facilitate the investigation of sudden changes by change animation.

For performing a comparison task, each row of the animated small multiples contains a user-selected seizure type in a specific time interval. A seizure propagation with a rhythmic type has a unique color in each row (branch). Each branch shows the sequential changes in a seizure type through time. The new branch will be added vertically to the animated small multiples layout (Fig. 5). The user can compare two different branches of animated multiples arranged vertically.

The comparison is not limited to branches; every two consecutive multiples can be compared in a row. A multiple shows a seizure’s type propagation recorded by electrodes at a specific time point (Fig. 5). The animated small multiples visualizes time points of

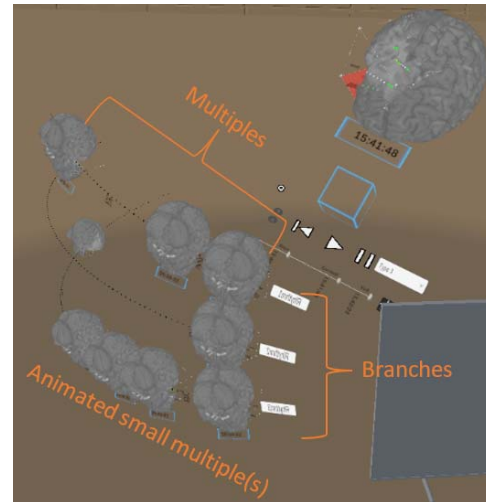


Figure 5: A view of RealityFlow demonstrating the arrangement of multiples and branches in the animated small multiples.

interest with different seizure patterns compared to a common small multiples visualization visualizing all time points. In fact, in a multiple of animated small multiples, at least a contact point exists whose status has changed from recording a seizure to not recording or vice versa compared to its previous neighbour multiple. The spatial distance between two multiples directly relates to the temporal distance between these multiples. Each bar shows the duration between two multiples (Fig. 4). Each dot in the bar corresponds to a second. To improve the investigation of changes, a change animation shows differences between two consecutive multiples when a user hovers over the latter multiple.

Another feature is proposed to accelerate the investigation of different time intervals in the central view. Instead of a back-and-forth investigation of a central view to find different time intervals or time points of a seizure spread, a user can control the displayed time in the central view by interacting with animated small multiples. When clicking on a multiple in a branch of animated small multiples, the central view time point will change to the time point of the selected multiple in the branch. The time interval of animation in the central view will also change to the time interval of the branch of the selected multiple.

4 CONCLUSION & FUTURE WORK

RealityFlow is a VR system to help neurologists in epilepsy surgical planning rounds. RealityFlow benefits from a VR environment's space to combine two popular visualization techniques, animation and small multiples. Our new technique visualizes seizure propagation dynamics in 3D brain models that can be explored by neurologists in virtual reality to better comprehend seizure onset, spread, and types important for surgical planning. In the future, we will extend RealityFlow to receive other brain imaging data for surgical planning, which can increase our system's usability for users and neurologists. Also, we will conduct user studies with the participation of neurologists to evaluate RealityFlow. These studies will help us to gain valuable insights and feedback from medical experts to help us better understand how RealityFlow can be integrated into the clinical workflow.

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